

Contraction Mechanisms for Generalized Sinkhorn Maps under ϕ -Divergence Regularization

A technical note

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Abstract

Entropic optimal transport has two useful order properties: the soft c -transform is order reversing and commutes with additive constants, hence the double transform is non-expansive in the variation seminorm. In the Kullback–Leibler case, strict contraction follows in compact bounded-cost settings from Hilbert’s projective metric and Birkhoff’s contraction theorem; Carlier’s proof gives another route through boundedness of Sinkhorn iterates, strong convexity of the exponential on bounded intervals, and a Gauss–Seidel/coordinate-descent comparison. This note explains which parts survive when the KL penalty is replaced by a general ϕ -divergence.

The main message is that monotonicity and non-expansiveness are structural, but strict contraction is not. For a smooth interior generalized transform, its derivative is a Markov averaging operator whose transition probabilities are proportional to the curvature of the dual integrand $\Phi = \varepsilon\phi_+^*(\cdot/\varepsilon)$. Therefore the correct replacement for positivity of the Gibbs kernel is a uniform overlap, or Doeblin, condition on these curvature kernels. A simple sufficient condition is a uniform strip on which $0 < m \leq \Phi'' \leq M < \infty$. The Carlier argument extends by replacing the exponential inequalities with the corresponding strong-convexity and Lipschitz estimates for Φ on that strip. Without such an interior curvature condition, sparse divergences such as quadratic/chi-square regularization can have local contraction factor one.

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1 Introduction

Let (X, α) and (Y, β) be probability spaces and let $c : X \times Y \rightarrow \mathbb{R}$ be a measurable cost. For a convex function $\phi : [0, \infty) \rightarrow (-\infty, +\infty]$ with $\phi(1) = 0$, the ϕ -regularized transport problem is

$$\inf_{\pi \in \Pi(\alpha, \beta)} \left\{ \int_{X \times Y} c \, d\pi + \varepsilon D_\phi(\pi \mid \alpha \otimes \beta) \right\}, \quad \varepsilon > 0. \quad (1)$$

When $\phi(r) = r \log r - r + 1$, this is the usual entropy-regularized optimal transport problem. Replacing KL by a general ϕ leads to a generalized Sinkhorn algorithm: one alternately maximizes the dual in one potential while keeping the other fixed. This generalized transform was introduced and studied in the f -divergence OT literature; in particular, Terjek and Gonzalez-Sanchez formulate the (c, ε, ϕ) -transform, its implicit equation, and convergence conditions for the generalized Sinkhorn algorithm [14]. Earlier convergence results for general convex regularization also appear in Di Marino–Gerolin [5].

For KL, several contraction analyses are available. The classical two-marginal compact proof uses Hilbert’s projective metric and Birkhoff–Hopf contraction for positive kernel operators; Carlier instead proves linear convergence of the multi-marginal Sinkhorn algorithm by combining boundedness of the iterates, strong convexity and Lipschitz continuity of the exponential on bounded intervals, and coordinate-descent estimates [3]. Recent work extends Hilbert-metric methods for KL to certain unbounded settings by changing the underlying cone and allowing kernels that are not bounded away from zero [7]. These facts suggest the question studied here:

Which contraction mechanisms extend from KL to a general ϕ -divergence?

The answer is nuanced. The order-theoretic non-expansiveness extends almost verbatim. Strict contraction requires an additional quantitative interiority condition. The most transparent way to see this is to differentiate the generalized transform. Its derivative is a Markov operator with kernel proportional to Φ'' , where

$$\Phi(s) := \varepsilon \phi_+^*(s/\varepsilon), \quad \phi_+(r) := \phi(r) + \iota_{[0, \infty)}(r). \quad (2)$$

Thus strict variation contraction is equivalent, at the infinitesimal level, to a Dobrushin overlap condition for those curvature kernels. This is automatic for KL on compact bounded-cost problems, but it can fail for other divergences. A two-point chi-square example in Section 7 shows that one cannot hope for a theorem saying that all generalized Sinkhorn maps are strict contractions.

The note is organized as follows. Section 2 recalls the dual and the generalized soft c -transform. Section 3 proves order reversal, translation covariance and non-expansiveness in the variation seminorm. Section 4 derives the derivative kernel and the Dobrushin contraction criterion. Section 5 explains why the KL case has additional Hilbert-metric structure. Section 6 formulates the Carlier-style extension through local strong convexity and smoothness of Φ . Section 7 gives the obstruction. The final section sketches weighted/noncompact extensions.

2 Dual problem and generalized soft c -transform

2.1 Normalization of the divergence

The affine part of ϕ does not affect the value of $D_\phi(\pi \mid \alpha \otimes \beta)$ on probability couplings: adding $a(r-1)$ to $\phi(r)$ integrates to zero when π and $\alpha \otimes \beta$ have the same total mass. We therefore use the following normalization throughout the note.

Assumption 2.1 (standing convex-analytic assumptions). *The function $\phi : [0, \infty) \rightarrow (-\infty, +\infty]$ is proper, lower semicontinuous and convex, with*

$$\phi(1) = 0, \quad 0 \in \partial\phi(1).$$

Let $\phi_+ = \phi + \iota_{[0, \infty)}$ and let $\Phi(s) = \varepsilon\phi_+^*(s/\varepsilon)$. On intervals where differentiability is needed, assume $\Phi \in C^2(I)$ and write

$$\vartheta(s) := \Phi'(s) = (\phi_+^*)'(s/\varepsilon).$$

The normalization gives $\vartheta(0) = 1$ whenever ϑ is defined at zero.

The usual generalized f -divergence includes a singular part:

$$D_\phi(\pi \mid m) = \int \phi\left(\frac{d\pi}{dm}\right) dm + \phi'_\infty \pi^\perp(\mathsf{X} \times \mathsf{Y}), \quad m = \alpha \otimes \beta,$$

where $\pi = \frac{d\pi}{dm}m + \pi^\perp$ is the Lebesgue decomposition and $\phi'_\infty = \lim_{r \rightarrow \infty} \phi(r)/r$. If $\phi'_\infty = +\infty$, singular mass is forbidden. If $\phi'_\infty < \infty$, the dual has a corresponding upper constraint. This is one reason why non-superlinear regularizations may partially collapse to the ordinary hard c -transform [14].

2.2 Fenchel dual

Formally, and rigorously under the usual compactness/continuity or integrable lower-bounded hypotheses, the dual of (1) is

$$\sup_{f, g} \left\{ \int_{\mathsf{X}} f \, d\alpha + \int_{\mathsf{Y}} g \, d\beta - \int_{\mathsf{X} \times \mathsf{Y}} \Phi(f(x) + g(y) - c(x, y)) \, d\alpha(x) \, d\beta(y) \right\}. \quad (3)$$

Equivalently, if one does not absorb the effective domain of ϕ_+^* into Φ , then $f(x) + g(y) - c(x, y) \leq \varepsilon\phi'_\infty$ is an explicit constraint. For KL, $\phi(r) = r \log r - r + 1$ and

$$\phi_+^*(t) = e^t - 1, \quad \Phi(s) = \varepsilon(e^{s/\varepsilon} - 1), \quad \vartheta(s) = e^{s/\varepsilon}.$$

The additive constant $-\varepsilon$ in Φ is irrelevant for optimization but convenient for normalization.

2.3 The generalized transform

Fix a potential f on X . The update of the Y -potential is the pointwise maximizer of the dual in $g(y)$:

$$T_\alpha f(y) := \arg \max_{\gamma} \left\{ \gamma - \int_X \Phi(f(x) + \gamma - c(x, y)) \, d\alpha(x) \right\}. \quad (4)$$

When the maximizer is interior and unique, it is characterized by

$$\int_X \vartheta(f(x) + T_\alpha f(y) - c(x, y)) \, d\alpha(x) = 1. \quad (5)$$

This is the generalized soft c -transform. The analogous map T_β sends functions on Y to functions on X :

$$\int_Y \vartheta(T_\beta g(x) + g(y) - c(x, y)) \, d\beta(y) = 1. \quad (6)$$

The generalized Sinkhorn map on X -potentials is

$$S := T_\beta T_\alpha. \quad (7)$$

Remark 2.2 (boundary cases). *If Φ is finite only on a half-line or if ϑ has a finite plateau, Equation (5) may fail to have an interior solution. Then the maximizer in (4) occurs at the boundary and the update becomes a hard c -transform plus the endpoint constant. Terjek–Gonzalez–Sanchez call out precisely this phenomenon in the non-superlinear case [14]. All differential contraction statements below are made on the interior region where (5) is valid.*

3 Order structure and variation non-expansiveness

For a bounded measurable function h on a probability space, define the variation seminorm

$$\|h\|_{\text{var}} := \text{osc}(h) := \text{ess sup } h - \text{ess inf } h.$$

This is the quotient norm on $L^\infty/\mathbb{R}$.

Proposition 3.1 (order reversal and translations). *Assume the transform T_α is single-valued. Then*

$$f \leq \tilde{f} \implies T_\alpha \tilde{f} \leq T_\alpha f, \quad (8)$$

$$T_\alpha(f + a) = T_\alpha f - a, \quad a \in \mathbb{R}. \quad (9)$$

Consequently $S = T_\beta T_\alpha$ is order preserving and satisfies $S(f + a) = Sf + a$.

Proof. The translation identity follows directly from (4) or (5). For order reversal, suppose $f \leq \tilde{f}$. For a fixed y , the function

$$\gamma \mapsto \int_X \vartheta(f(x) + \gamma - c(x, y)) \, d\alpha(x)$$

is nondecreasing in both f and γ . If $\gamma_f = T_\alpha f(y)$ solves (5), then at the same value γ_f the corresponding integral for $\tilde{f}$ is at least one. To make it equal to one, the value of γ must be no larger. The same argument can be written at the level of the concave maximization problem (4); in nonsmooth cases one uses monotonicity of the subdifferential of Φ . $\square$

Proposition 3.2 (non-expansiveness in variation). *Let T be an order-reversing map with $T(f + a) = Tf - a$. Then*

$$\text{osc}(Tf - T\tilde{f}) \leq \text{osc}(f - \tilde{f}). \quad (10)$$

If S is order preserving and $S(f + a) = Sf + a$, then

$$\text{osc}(Sf - S\tilde{f}) \leq \text{osc}(f - \tilde{f}). \quad (11)$$

In particular, the generalized double transform is non-expansive in $L^\infty/\mathbb{R}$ whenever the transforms are single-valued.

Proof. Let

$$m := \text{ess inf}(f - \tilde{f}), \quad M := \text{ess sup}(f - \tilde{f}).$$

Then $\tilde{f} + m \leq f \leq \tilde{f} + M$. Applying an order-reversing T and using translation covariance gives

$$T\tilde{f} - M \leq Tf \leq T\tilde{f} - m.$$

Therefore $Tf - T\tilde{f} \in [-M, -m]$ a.e., proving (10). The order-preserving case is identical, with $S\tilde{f} + m \leq Sf \leq S\tilde{f} + M$. $\square$

Remark 3.3. *This argument gives non-expansiveness only. It contains no mechanism forcing a strict improvement of the oscillation. Strictness depends on how much averaging is performed by the transform. The next section identifies the averaging kernel.*

4 Derivative kernels and Dobrushin contraction

4.1 Derivative of a generalized transform

Assume for this section that the update is interior, that $\Phi \in C^2$ on the range of all quantities involved, and that

$$\int_{\mathbf{X}} \Phi''(f(x) + T_\alpha f(y) - c(x, y)) \, d\alpha(x) > 0$$

for every y . Put

$$z_y^f(x) := f(x) + T_\alpha f(y) - c(x, y). \quad (12)$$

Proposition 4.1 (linearization). *The Frechet derivative of T_α at f is*

$$DT_\alpha(f)[h](y) = - \int_{\mathbf{X}} h(x) P_y^f(\,dx), \quad (13)$$

where P_y^f is the probability measure

$$P_y^f(\,dx) = p_y^f(x) \alpha(\,dx), \quad p_y^f(x) = \frac{\Phi''(z_y^f(x))}{\int_{\mathbf{X}} \Phi''(z_y^f(x')) \, d\alpha(x')}. \quad (14)$$

Proof. Differentiate the identity

$$\int_{\mathbf{X}} \Phi'(f(x) + T_\alpha f(y) - c(x, y)) \, d\alpha(x) = 1$$

in the direction h . This gives

$$\int_{\mathbf{X}} \Phi''(z_y^f(x)) (h(x) + DT_\alpha(f)[h](y)) \, d\alpha(x) = 0,$$

which is exactly (13). $\square$

Thus the derivative of T_α is the negative of a Markov averaging operator. The minus sign is harmless for oscillations.

4.2 Dobrushin coefficient

For a Markov kernel $P = (P_y)_{y \in Y}$ from Y to X , define its Dobrushin coefficient

$$\delta(P) := \sup_{y, y'} \|P_y - P_{y'}\|_{TV} = 1 - \inf_{y, y'} \int_X \min(p_y, p_{y'}) d\alpha, \quad (15)$$

whenever $P_y = p_y \alpha$. Here $\|\mu - \nu\|_{TV} = \sup_A |\mu(A) - \nu(A)| = \frac{1}{2} \int |p - q| d\alpha$. The standard Dobrushin estimate says

$$\text{osc} \left(y \mapsto \int h dP_y \right) \leq \delta(P) \text{osc}(h). \quad (16)$$

Combining this estimate with Proposition 4.1 gives the infinitesimal contraction factor of the generalized transform.

Corollary 4.2 (local variation contraction). *At an interior smooth point f ,*

$$\text{osc}(DT_\alpha(f)[h]) \leq \delta(P^f) \text{osc}(h), \quad (17)$$

where P^f is the curvature kernel (14). In particular, T_α is strictly contracting at f only if the family $(P_y^f)_{y \in Y}$ has nontrivial overlap:

$$\lambda(f) := \inf_{y, y'} \int_X \min(p_y^f, p_{y'}^f) d\alpha > 0. \quad (18)$$

The corresponding infinitesimal factor is at most $1 - \lambda(f)$.

4.3 A nonlinear contraction criterion

The previous statement is infinitesimal. A uniform estimate along line segments gives a genuine Lipschitz contraction.

Theorem 4.3 (Dobrushin criterion for the generalized transform). *Let $\mathcal{U}$ be a convex class of potentials on X such that T_α is interior and C^1 on $\mathcal{U}$. Assume that*

$$\delta(P^f) \leq \kappa_X < 1, \quad f \in \mathcal{U}. \quad (19)$$

Then

$$\text{osc}(T_\alpha f - T_\alpha \tilde{f}) \leq \kappa_X \text{osc}(f - \tilde{f}), \quad f, \tilde{f} \in \mathcal{U}. \quad (20)$$

If, similarly, along the corresponding Y -potentials one has

$$\delta(Q^g) \leq \kappa_Y < 1$$

for the derivative kernels of T_β , then the double transform satisfies

$$\text{osc}(Sf - S\tilde{f}) \leq \kappa_X \kappa_Y \text{osc}(f - \tilde{f}). \quad (21)$$

Proof. For $f_t = (1 - t)f + t\tilde{f}$,

$$T_\alpha \tilde{f} - T_\alpha f = \int_0^1 DT_\alpha(f_t)[\tilde{f} - f] dt.$$

Using (17) and (19) yields (20). The estimate for $S = T_\beta T_\alpha$ follows by applying the one-block estimate twice. $\square$

4.4 Uniform curvature strip

The Dobrushin overlap condition is conceptually exact but may be hard to check. A crude, useful sufficient condition is uniform curvature on a strip.

Corollary 4.4 (curvature strip implies contraction). *Assume that for every potential under consideration and every x, y ,*

$$z_y^f(x) = f(x) + T_\alpha f(y) - c(x, y) \in I, \quad (22)$$

where $I \subset \mathbb{R}$ is an interval on which

$$0 < \mu_I := \inf_{s \in I} \Phi''(s) \leq L_I := \sup_{s \in I} \Phi''(s) < \infty. \quad (23)$$

Then

$$\text{osc}(T_\alpha f - T_\alpha \tilde{f}) \leq \left(1 - \frac{\mu_I}{L_I}\right) \text{osc}(f - \tilde{f}) \quad (24)$$

provided the same strip condition holds along the segment between f and $\tilde{f}$.

Proof. For each y ,

$$\int_{\mathbf{x}} \Phi''(z_y^f(x)) \, d\alpha(x) \leq L_I,$$

while $\Phi''(z_y^f(x)) \geq \mu_I$. Hence the density p_y^f in (14) satisfies $p_y^f(x) \geq \mu_I/L_I$ for α -a.e. x . Therefore

$$\int \min(p_y^f, p_{y'}^f) \, d\alpha \geq \mu_I/L_I,$$

and Theorem 4.3 applies. $\square$

Remark 4.5 (the strip estimate is not sharp). *For KL, $\Phi''(s) = \varepsilon^{-1}e^{s/\varepsilon}$. On a strip of width R one gets the rough factor $1 - e^{-R/\varepsilon}$. Hilbert-metric estimates exploit the special multiplicative structure of the exponential and typically produce sharper or more intrinsic constants. For a general ϕ , however, the curvature-kernel criterion is the natural replacement.*

5 What is special about KL?

For KL,

$$\vartheta(s) = e^{s/\varepsilon}.$$

The optimal density associated with dual potentials is

$$\rho(x, y) = e^{(f(x)+g(y)-c(x,y))/\varepsilon} = a(x)K(x, y)b(y), \quad (25)$$

where

$$a(x) = e^{f(x)/\varepsilon}, \quad b(y) = e^{g(y)/\varepsilon}, \quad K(x, y) = e^{-c(x,y)/\varepsilon}.$$

Sinkhorn updates are therefore positive linear kernel operations followed by diagonal rescalings. Hilbert's projective metric is built precisely to study positive homogeneous maps on cones. If K is bounded above and below on a compact domain, Birkhoff–Hopf gives strict projective contraction. Recent work extends this strategy for KL by changing the cone and allowing controlled decay of K in unbounded settings [7].

For a general divergence,

$$\rho(x, y) = \vartheta(f(x) + g(y) - c(x, y)). \quad (26)$$

There is no factorization of the form $a(x)K(x, y)b(y)$ unless ϑ is exponential. Indeed, a positive function satisfying $\vartheta(r + s) = A(r)B(s)$, under minimal regularity, is exponential up to constants. Thus the direct Hilbert-metric proof is a KL-specific argument.

The derivative formula shows what remains. Even though the nonlinear map is no longer a projective positive linear map, its linearization is still a Markov averaging operator, now with weights proportional to Φ'' . Consequently:

$$\text{KL Hilbert positivity} \rightsquigarrow \text{uniform overlap of curvature kernels for general } \phi.$$

This is the main structural replacement.

6 Carrier-style extension through strong convexity

Carrier's proof of linear convergence for entropic multi-marginal Sinkhorn is based on three ingredients: bounded Sinkhorn iterates, strong convexity of the exponential on the bounded range reached by the iterates, and a standard cyclic-coordinate-descent comparison [3]. This mechanism is not specific to the exponential. It extends to any Φ satisfying uniform strong convexity and smoothness on the relevant range.

6.1 Multi-marginal dual functional

Let (X_i, m_i) , $i = 1, \dots, N$, be probability spaces and let $m = m_1 \otimes \dots \otimes m_N$. Write $x = (x_1, \dots, x_N)$ and let $c : \prod_i X_i \rightarrow \mathbb{R}$. For potentials $u = (u_1, \dots, u_N)$, define

$$F(u) := \int_{\prod_i X_i} \Phi \left(\sum_{i=1}^N u_i(x_i) - c(x) \right) dm(x) - \sum_{i=1}^N \int_{X_i} u_i dm_i. \quad (27)$$

Maximizing the dual is equivalent to minimizing F . The functional is invariant under adding constants a_i to the blocks whenever $\sum_i a_i = 0$, so one works on a gauge-fixed affine space, for example

$$E = \left\{ u : \int u_i dm_i = 0 \text{ for } i = 1, \dots, N-1 \right\}.$$

Cyclic generalized Sinkhorn is exact block minimization of F over $u_1, u_2, \dots, u_N$, repeated.

The block optimality condition for the i -th update is

$$\int_{\prod_{j \neq i} X_j} \vartheta \left(u_i(x_i) + \sum_{j \neq i} u_j(x_j) - c(x) \right) dm_{-i}(x_{-i}) = 1 \quad (28)$$

for m_i -a.e. x_i , with the other blocks fixed. For $N = 2$ this is exactly (5) or (6).

6.2 The two scalar inequalities that replace KL

Let $I \subset \mathbb{R}$ be an interval and suppose

$$0 < \mu_I \leq \Phi''(s) \leq L_I < \infty, \quad s \in I. \quad (29)$$

Then for all $a, b \in I$,

$$\Phi(b) - \Phi(a) - \Phi'(a)(b - a) \geq \frac{\mu_I}{2}(b - a)^2, \quad (30)$$

$$|\Phi'(b) - \Phi'(a)| \leq L_I |b - a|. \quad (31)$$

In Carrier's KL proof these are used with $\Phi(s) = e^s$ on a bounded interval:

$$e^b - e^a - e^a(b - a) \geq \frac{e^{-M}}{2}(b - a)^2, \quad |e^b - e^a| \leq e^M |b - a|, \quad a, b \in [-M, M].$$

Thus (30)–(31) are the only replacements needed at the scalar level.

6.3 A reusable sufficient theorem

The following theorem is deliberately stated as a sufficient criterion, not as an optimal convergence theorem. Its purpose is to isolate what must be checked for a given divergence and a given class of costs.

Theorem 6.1 (Carrier-type linear convergence for general ϕ). *Consider the functional F in (27) on a gauge-fixed space E . Assume:*

- (A1) *The cyclic block minimizers are well-defined and remain in an interior region.*
- (A2) *There exists an interval I such that, for every iterate u^t , every intermediate Gauss–Seidel state in the cycle, and an optimal potential u^* ,*

$$\sum_{i=1}^N u_i(x_i) - c(x) \in I \quad (32)$$

for m-a.e. x .

- (A3) *On I , Φ satisfies $0 < \mu_I \leq \Phi'' \leq L_I < \infty$.*

- (A4) *On the chosen gauge E , the minimizer u^* is unique.*

Then the generalized Sinkhorn/Gauss–Seidel iterates converge to u^ in L^2 block norm, and the objective values converge linearly. More precisely, one obtains a non-optimized estimate of the form*

$$F(u^t) - F(u^*) \leq \left(1 - \frac{1}{N} \left(\frac{\mu_I}{L_I}\right)^2\right)^t (F(u^0) - F(u^*)), \quad (33)$$

up to harmless changes of the numerical constant depending on the precise gauge and block norm convention.

Proof sketch. Let $u^{t,i}$ be the intermediate state after blocks $1, \dots, i$ have been updated during cycle t , with $u^{t,0} = u^t$ and $u^{t,N} = u^{t+1}$. Since block i is exactly minimized, the first variation in that block vanishes at $u^{t,i}$. Applying (30) to the change from $u^{t,i-1}$ to $u^{t,i}$ and summing over the blocks yields the sufficient-decrease estimate

$$F(u^t) - F(u^{t+1}) \geq \frac{\mu_I}{2} \sum_{i=1}^N \|u_i^{t+1} - u_i^t\|_{L^2(m_i)}^2. \quad (34)$$

This is the direct analogue of Carrier’s Lemma 3.2.

To compare the current gap with the step size, use convexity at u^t and insert intermediate states $u^{t,i}$ for which the block gradients vanish. The difference between the gradient at u^t and the gradient at $u^{t,i}$ is controlled by (31); Young’s inequality and (30) give an estimate of the type

$$F(u^t) - F(u^*) \leq \frac{NL_I^2}{2\mu_I} \sum_{i=1}^N \|u_i^{t+1} - u_i^t\|_{L^2(m_i)}^2. \quad (35)$$

Combining (34) and (35) gives

$$F(u^t) - F(u^{t+1}) \geq \frac{1}{N} \left(\frac{\mu_I}{L_I}\right)^2 (F(u^t) - F(u^*)),$$

which is (33). The L^2 convergence follows from summing (34), compactness or weak compactness of bounded iterates, and uniqueness on the gauge-fixed quotient, as in the KL proof. $\square$

6.4 How to obtain the strip

For bounded costs and normalized $\vartheta(0) = 1$, the transform itself gives useful oscillation estimates. Fix y and write $a_y(x) = f(x) - c(x, y)$. Equation (5) is

$$\int \vartheta(a_y(x) + \gamma) d\alpha(x) = 1.$$

Since ϑ is increasing and $\vartheta(0) = 1$, the solution satisfies

$$-\operatorname{ess\,sup}_x a_y(x) \leq \gamma \leq -\operatorname{ess\,inf}_x a_y(x). \quad (36)$$

Equivalently,

$$z_y^f(x) = a_y(x) + \gamma \in [-\operatorname{osc}(a_y), \operatorname{osc}(a_y)] \quad \text{a.e.} \quad (37)$$

Moreover, comparing two points y, y' gives

$$|T_\alpha f(y) - T_\alpha f(y')| \leq \operatorname{ess\,sup}_x |c(x, y) - c(x, y')|. \quad (38)$$

Thus after one block update the oscillation of the updated potential is controlled by the oscillation of the cost, independently of the incoming potential. In bounded-cost settings this usually gives bounded potentials after fixing a gauge, hence a bounded interval containing the values of $\sum_i u_i - c$.

This bounded interval is not yet enough: it must lie in a region where Φ'' is bounded above and below. KL satisfies this on every bounded interval. Many other divergences satisfy it only in a proper interior region. For sparse regularizers, the lower bound may fail because Φ'' vanishes on inactive parts of the domain.

7 A counterexample: chi-square can have contraction factor one

The obstruction is not merely technical. Some ϕ -divergences generate sparse couplings, and sparse active sets can destroy strict contraction.

Consider

$$\phi(r) = \frac{1}{2}(r-1)^2, \quad r \geq 0.$$

Then

$$\phi_+^*(t) = \begin{cases} t + \frac{1}{2}t^2, & t \geq -1, \\ -\frac{1}{2}, & t < -1, \end{cases} \quad \vartheta(s) = \Phi'(s) = \left(1 + \frac{s}{\varepsilon}\right)_+,$$

so

$$\Phi''(s) = \frac{1}{\varepsilon} \mathbf{1}_{\{s > -\varepsilon\}}$$

away from the kink. The curvature kernel is uniform on the active set $\{s > -\varepsilon\}$ and zero outside it.

Let

$$\mathsf{X} = \mathsf{Y} = \{1, 2\}, \quad \alpha = \beta = \left(\frac{1}{2}, \frac{1}{2}\right),$$

and choose the cost matrix

$$c = \begin{pmatrix} 0 & L \\ L & 0 \end{pmatrix}, \quad L > 2\varepsilon. \quad (39)$$

Start from $f = (0, 0)$. For $y = 1$, the update $g_1 = T_\alpha f(1)$ solves

$$\frac{1}{2} \left(1 + \frac{g_1}{\varepsilon}\right)_+ + \frac{1}{2} \left(1 + \frac{g_1 - L}{\varepsilon}\right)_+ = 1.$$

Because $L > 2\varepsilon$, the solution is $g_1 = \varepsilon$, and only $x = 1$ is active. Similarly $g_2 = \varepsilon$, and only $x = 2$ is active. Hence the derivative kernels are

$$P_1 = \delta_1, \quad P_2 = \delta_2.$$

By Proposition 4.1,

$$DT_\alpha(0)[h] = (-h_1, -h_2).$$

Applying the second block update gives the same derivative structure, hence

$$D(T_\beta T_\alpha)(0) = \text{id}.$$

The local contraction factor of the double transform in variation seminorm is therefore exactly one.

Remark 7.1. *This example shows that strict contraction cannot be true for arbitrary ϕ -divergences without an overlap/interiority assumption. It also explains the mechanism: different target points see disjoint active source sets, so the Markov kernels P_y have zero common mass.*

8 Noncompact and weighted extensions

The compact bounded-cost theory obtains strict contraction from a uniform minorization: all derivative kernels share a fixed amount of mass. On noncompact spaces with unbounded costs, this is usually false globally. There are two natural ways to proceed.

8.1 Weighted variation and Harris-type estimates

Replace the oscillation seminorm by a weighted quotient seminorm, for example

$$\|h\|_{V, \text{var}} := \inf_{a \in \mathbb{R}} \left\| \frac{h - a}{V} \right\|_\infty, \quad V \geq 1. \quad (40)$$

The derivative kernels P_y^f then need not satisfy a global Doeblin condition. It is enough to prove a Harris-type combination of

- (i) a minorization on a large set where the cost is controlled and Φ'' is bounded away from zero;
- (ii) a drift estimate showing that P_y^f does not place too much mass in the tails relative to V ;
- (iii) a priori weighted bounds for the potentials and for $f \oplus g - c$.

This is the natural general- ϕ analogue of the way KL Hilbert-metric arguments have been adapted to unbounded kernels by modifying the cone and controlling tail growth [7]. The difference is that the kernel to control is not the Gibbs kernel $e^{-c/\varepsilon}$ itself, but the curvature kernel $\Phi''(f \oplus g - c)$ normalized in each fiber.

8.2 Weighted Carlier estimates

The Carlier route asks for a different set of estimates. One needs weighted versions of

$$\mu_I \leq \Phi'' \leq L_I$$

not necessarily on a bounded interval, but on the region reached by the iterates. A typical assumption would be that for some weights V_i ,

$$|u_i^t(x_i)| \lesssim V_i(x_i), \quad \sum_i u_i^t(x_i) - c(x) \in I(x)$$

with

$$0 < \mu(x) \leq \Phi''(s) \leq L(x) < \infty, \quad s \in I(x),$$

and with the weighted ratios integrable enough to reproduce the two scalar estimates (30)–(31) in a weighted L^2 norm. This is plausible for divergences whose curvature behaves regularly in the tails, but it is no longer automatic. The key task is always the same: prevent the iterates from approaching a region where Φ'' either vanishes or explodes too fast.

9 Checklist for extending a KL proof to a given divergence

Given a specific ϕ , the following checklist isolates the work needed to obtain strict contraction or linear convergence.

Step 1. Dual integrand. Compute $\Phi(s) = \varepsilon \phi_+^*(s/\varepsilon)$, its derivative $\vartheta = \Phi'$, and its curvature Φ'' on the interior of its domain.

Step 2. Interior transform. Show that the block equation

$$\int \vartheta(f + \gamma - c) d\alpha = 1$$

has a unique interior solution along the iterates. If the solution can hit the boundary, strict contraction may fail.

Step 3. A priori range. Prove that $f \oplus g - c$ remains in a region I or in a weighted region where the curvature is controlled.

Step 4. Variation contraction route. Verify a uniform overlap condition for the curvature kernels

$$P_y^f(dx) \propto \Phi''(f(x) + T_\alpha f(y) - c(x, y)) \alpha(dx).$$

A sufficient condition is $0 < \inf_I \Phi'' \leq \sup_I \Phi'' < \infty$.

Step 5. L^2 coordinate-descent route. Verify the scalar inequalities (30)–(31) on the same region and reproduce the sufficient-decrease and gap-to-step estimates.

Step 6. Gauge and compactness. Fix the additive constants and prove convergence of subsequences or compactness in the relevant topology. Strict convexity on the quotient then identifies the limit.

10 Conclusion

The order-theoretic part of Sinkhorn is robust. For a broad class of ϕ -divergences, the generalized transform is order reversing and translation covariant; the double transform is therefore non-expansive in the variation seminorm. Strict contraction is a different matter. In KL it is supported by the exponential factorization, which makes the algorithm a projective positive-kernel iteration. General ϕ -regularization loses this multiplicative structure.

The replacement principle is:

strict contraction comes from overlap of the curvature kernels

with kernels proportional to $\Phi''(f \oplus g - c)$. If the iterates stay in an interior strip where $0 < \mu \leq \Phi'' \leq L < \infty$, then the variation and Carlier-style L^2 analyses extend with constants depending on μ/L . If that interiority fails, sparse divergences can have disjoint active sets and local contraction factor one.

References

- [1] A. Beck and L. Tetruashvili. On the convergence of block coordinate descent type methods. *SIAM Journal on Optimization*, 23(4):2037–2060, 2013.
- [2] G. Birkhoff. Extensions of Jentzsch’s theorem. *Transactions of the American Mathematical Society*, 85(1):219–227, 1957.
- [3] G. Carlier. On the linear convergence of the multi-marginal Sinkhorn algorithm. *SIAM Journal on Optimization*, 32(2):786–794, 2022. Preprint version: <https://mathtube.org/sites/default/files/lecture-extra-files/linear-sinkhorn.pdf>.
- [4] M. Cuturi. Sinkhorn distances: Lightspeed computation of optimal transport. In *Advances in Neural Information Processing Systems*, 2013.
- [5] S. Di Marino and A. Gerolin. Optimal transport losses and Sinkhorn algorithm with general convex regularization. *arXiv:2007.00976*, 2020.
- [6] R. L. Dobrushin. Central limit theorem for nonstationary Markov chains. I, II. *Theory of Probability and its Applications*, 1:65–80, 329–383, 1956.
- [7] S. Eckstein. Hilbert’s projective metric for functions of bounded growth and exponential convergence of Sinkhorn’s algorithm. *Probability Theory and Related Fields*, 2025. See also *arXiv:2311.04041*.
- [8] S. P. Eveson and R. D. Nussbaum. An elementary proof of the Birkhoff–Hopf theorem. *Mathematical Proceedings of the Cambridge Philosophical Society*, 117(1):31–55, 1995.
- [9] J. Franklin and J. Lorenz. On the scaling of multidimensional matrices. *Linear Algebra and its Applications*, 114/115:717–735, 1989.
- [10] P. Ghosal and M. Nutz. On the convergence rate of Sinkhorn’s algorithm. Preprint, 2025. https://www.math.columbia.edu/~mnutz/docs/Sinkhorn_rate.pdf.
- [11] M. Nutz. Introduction to entropic optimal transport. Lecture notes, 2022. https://www.math.columbia.edu/~mnutz/docs/EOT_lecture_notes.pdf.
- [12] G. Peyre and M. Cuturi. Computational optimal transport. *Foundations and Trends in Machine Learning*, 11(5–6):355–607, 2019.
- [13] R. Sinkhorn and P. Knopp. Concerning nonnegative matrices and doubly stochastic matrices. *Pacific Journal of Mathematics*, 21(2):343–348, 1967.
- [14] D. Terjek and D. Gonzalez-Sanchez. Optimal transport with f -divergence regularization and generalized Sinkhorn algorithm. In *Proceedings of AISTATS 2022*, PMLR 151:5135–5165, 2022. <https://proceedings.mlr.press/v151/terjek22a.html>.
- [15] C. Villani. *Optimal Transport: Old and New*. Springer, 2009.